

## Chapter 1

# Accessing Systems and Obtaining Support

### Goal

Log in to local and remote Linux system, and investigate problem resolution methods provided through Red Hat Support and Red Hat Insights.

### Objectives

- Log in to a Linux system on a local text console and run simple commands using the shell.
- Configure key-based authentication for a user account to log in to remote systems securely without a password.
- Describe key resources available through the Red Hat Customer Portal, and find information from Red Hat documentation and the Knowledgebase.
- Analyze servers for issues, remediate or resolve them, and confirm the solution with Red Hat Insights.

### Sections

- Accessing the Command Line (and Quiz)
- Configuring SSH Key-Based Authentication (and Guided Exercise)
- Getting Help from Red Hat Customer Portal (and Guided Exercise)
- Detecting and Resolving Issues with Red Hat Insights (and Quiz)

# Accessing the Command Line

## Objectives

After completing this section, you should be able to log in to a Linux system and run simple commands using the shell.

## Introduction to the Bash Shell

A *command line* is a text-based interface which can be used to input instructions to a computer system. The Linux command line is provided by a program called the *shell*. Various options for the shell program have been developed over the years, and different users can be configured to use different shells. Most users, however, stick with the current default.

The default shell for users in Red Hat Enterprise Linux is the GNU Bourne-Again Shell (**bash**). Bash is an improved version of one of the most successful shells used on UNIX-like systems, the Bourne Shell (**sh**).

When a shell is used interactively, it displays a string when it is waiting for a command from the user. This is called the *shell prompt*. When a regular user starts a shell, the default prompt ends with a **\$** character, as shown below.

```
[user@host ~]$
```

The **\$** character is replaced by a **#** character if the shell is running as the superuser, **root**. This makes it more obvious that it is a superuser shell, which helps to avoid accidents and mistakes which can affect the whole system. The superuser shell prompt is shown below.

```
[root@host ~]#
```

Using **bash** to execute commands can be powerful. The **bash** shell provides a scripting language that can support automation of tasks. The shell has additional capabilities that can simplify or make possible operations that are hard to accomplish efficiently with graphical tools.



### Note

The **bash** shell is similar in concept to the command-line interpreter found in recent versions of Microsoft Windows, **cmd.exe**, although **bash** has a more sophisticated scripting language. It is also similar to Windows PowerShell in Windows 7 and Windows Server 2008 R2 and later. Administrators using the Apple Mac who use the Terminal utility may be pleased to note that **bash** is the default shell in macOS.

## Shell Basics

Commands entered at the shell prompt have three basic parts:

- *Command* to run
- *Options* to adjust the behavior of the command
- *Arguments*, which are typically targets of the command

The *command* is the name of the program to run. It may be followed by one or more *options*, which adjust the behavior of the command or what it will do. Options normally start with one or two dashes (**-a** or **--all**, for example) to distinguish them from arguments. Commands may also be followed by one or more *arguments*, which often indicate a target that the command should operate upon.

For example, the command **usermod -L user01** has a command (**usermod**), an option (**-L**), and an argument (**user01**). The effect of this command is to lock the password of the **user01** user account.

## Logging in to a Local Computer

To run the shell, you need to log in to the computer on a *terminal*. A terminal is a text-based interface used to enter commands into and print output from a computer system. There are several ways to do this.

The computer might have a hardware keyboard and display for input and output directly connected to it. This is the Linux machine's *physical console*. The physical console supports multiple *virtual consoles*, which can run separate terminals. Each virtual console supports an independent login session. You can switch between them by pressing **Ctrl+Alt** and a function key (**F1** through **F6**) at the same time. Most of these virtual consoles run a terminal providing a text login prompt, and if you enter your username and password correctly, you will log in and get a shell prompt.

The computer might provide a graphical login prompt on one of the virtual consoles. You can use this to log in to a *graphical environment*. The graphical environment also runs on a virtual console. To get a shell prompt you must start a terminal program in the graphical environment. The shell prompt is provided in an application window of your graphical terminal program.



### Note

Many system administrators choose not to run a graphical environment on their servers. This allows resources which would be used by the graphical environment to be used by the server's services instead.

In Red Hat Enterprise Linux 8, if the graphical environment is available, the login screen will run on the first virtual console, called **tty1**. Five additional text login prompts are available on virtual consoles two through six.

If you log in using the graphical login screen, your graphical environment will start on the first virtual console that is not currently being used by a login session. Normally, your graphical session will replace the login prompt on the second virtual console (**tty2**). However, if that console is in use by an active text login session (not just a login prompt), the next free virtual console is used instead.

The graphical login screen continues to run on the first virtual console (**tty1**). If you are already logged in to a graphical session, and log in as another user on the graphical login screen or use the **Switch User** menu item to switch users in the graphical environment without logging out, another graphical environment will be started for that user on the next free virtual console.

When you log out of a graphical environment, it will exit and the physical console will automatically switch back to the graphical login screen on the first virtual console.

**Note**

In Red Hat Enterprise Linux 6 and 7, the graphical login screen runs on the first virtual console, but when you log in your initial graphical environment *replaces* the login screen on the first virtual console instead of starting on a new virtual console.

In Red Hat Enterprise Linux 5 and earlier, the first six virtual consoles always provided text login prompts. If the graphical environment is running, it is on virtual console seven (accessed through **Ctrl+Alt+F7**).

A *headless server* does not have a keyboard and display permanently connected to it. A data center may be filled with many racks of headless servers, and not providing each with a keyboard and display saves space and expense. To allow administrators to log in, a headless server might have a login prompt provided by its *serial console*, running on a serial port which is connected to a networked console server for remote access to the serial console.

The serial console would normally be used to fix the server if its own network card became misconfigured and logging in over its own network connection became impossible. Most of the time, however, headless servers are accessed by other means over the network.

## Logging in over the Network

Linux users and administrators often need to get shell access to a remote system by connecting to it over the network. In a modern computing environment, many headless servers are actually virtual machines or are running as public or private cloud instances. These systems are not physical and do not have real hardware consoles. They might not even provide access to their (simulated) physical console or serial console.

In Linux, the most common way to get a shell prompt on a remote system is to use Secure Shell (SSH). Most Linux systems (including Red Hat Enterprise Linux) and macOS provide the OpenSSH command-line program **ssh** for this purpose.

In this example, a user with a shell prompt on the machine **host** uses **ssh** to log in to the remote Linux system **remotehost** as the user **remoteuser**:

```
[user@host ~]$ ssh remoteuser@remotehost
remoteuser@remotehost's password: password
[remoteuser@remotehost ~]$
```

The **ssh** command encrypts the connection to secure the communication against eavesdropping or hijacking of the passwords and content.

Some systems (such as new cloud instances) do not allow users to use a password to log in with **ssh** for tighter security. An alternative way to authenticate to a remote machine without entering a password is through *public key authentication*.

With this authentication method, users have a special identity file containing a *private key*, which is equivalent to a password, and which they keep secret. Their account on the server is configured with a matching *public key*, which does not have to be secret. When logging in, users can configure **ssh** to provide the private key and if their matching public key is installed in that account on that remote server, it will log them in without asking for a password.

In the next example, a user with a shell prompt on the machine **host** logs in to **remotehost** as **remoteuser** using **ssh**, using public key authentication. The **-i** option is used to specify the user's private key file, which is **mylab.pem**. The matching public key is already set up as an authorized key in the **remoteuser** account.

```
[user@host ~]$ ssh -i mylab.pem remoteuser@remotehost
[remoteuser@remotehost ~]$
```

For this to work, the private key file must be readable only by the user that owns the file. In the preceding example, where the private key is in the **mylab.pem** file, the command **chmod 600 mylab.pem** could be used to ensure this. How to set file permissions is discussed in more detail in a later chapter.

Users might also have private keys configured that are tried automatically, but that discussion is beyond the scope of this section. The References at the end of this section contain links to more information on this topic.



#### Note

The first time you log in to a new machine, you will be prompted with a warning from **ssh** that it cannot establish the authenticity of the host:

```
[user@host ~]$ ssh -i mylab.pem remoteuser@remotehost
The authenticity of host 'remotehost (192.0.2.42)' can't be established.
ECDSA key fingerprint is 47:bf:82:cd:fa:68:06:ee:d8:83:03:1a:bb:29:14:a3.
Are you sure you want to continue connecting (yes/no)? yes
[remoteuser@remotehost ~]$
```

Each time you connect to a remote host with **ssh**, the remote host sends **ssh** its *host key* to authenticate itself and to help set up encrypted communication. The **ssh** command compares that against a list of saved host keys to make sure it has not changed. If the host key has changed, this might indicate that someone is trying to pretend to be that host to hijack the connection which is also known as man-in-the-middle attack. In SSH, host keys protect against man-in-the-middle attacks, these host keys are unique for each server, and they need to be changed periodically and whenever a compromise is suspected.

You will get this warning if your local machine does not have a host key saved for the remote host. If you enter **yes**, the host key that the remote host sent will be accepted and saved for future reference. Login will continue, and you should not see this message again when connecting to this host. If you enter **no**, the host key will be rejected and the connection closed.

If the local machine does have a host key saved and it does not match the one actually sent by the remote host, the connection will automatically be closed with a warning.

## Logging Out

When you are finished using the shell and want to quit, you can choose one of several ways to end the session. You can enter the **exit** command to terminate the current shell session. Alternatively, finish a session by pressing **Ctrl+D**.

The following is an example of a user logging out of an SSH session:

```
[remoteuser@remotehost ~]$ exit
logout
Connection to remotehost closed.
[user@host ~]$
```



### References

**intro(1)**, **bash(1)**, **console(4)**, **pts(4)**, **ssh(1)**, and **ssh-keygen(1)** man pages

*Note: Some details of the **console(4)** man page, involving **init(8)** and **inittab(5)**, are outdated.*

For more information on OpenSSH and public key authentication, refer to the *Using secure communications between two systems with OpenSSH* chapter in the *Red Hat Enterprise Linux 8 Securing networks* guide at [https://access.redhat.com/documentation/en-us/red\\_hat\\_enterprise\\_linux/8/html-single/securing\\_networks/index#using-secure-communications-between-two-systems-with-openssh\\_securing-networks](https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/8/html-single/securing_networks/index#using-secure-communications-between-two-systems-with-openssh_securing-networks)



### Note

Instructions on how to read **man** pages and other online help documentation is included at the end of the next section.

## ► Quiz

# Accessing the Command Line

Choose the correct answer to the following questions:

- 1. Which term describes the interpreter that executes commands typed as strings?
  - a. Command
  - b. Console
  - c. Shell
  - d. Terminal
  
- 2. Which term describes the visual cue that indicates an interactive shell is waiting for the user to type a command?
  - a. Argument
  - b. Command
  - c. Option
  - d. Prompt
  
- 3. Which term describes the name of a program to run?
  - a. Argument
  - b. Command
  - c. Option
  - d. Prompt
  
- 4. Which term describes the part of the command line that adjusts the behavior of a command?
  - a. Argument
  - b. Command
  - c. Option
  - d. Prompt
  
- 5. Which term describes the part of the command line that specifies the target that the command should operate on?
  - a. Argument
  - b. Command
  - c. Option
  - d. Prompt

- ▶ **6. Which term describes the hardware display and keyboard used to interact with a system?**
  - a. Physical Console
  - b. Virtual Console
  - c. Shell
  - d. Terminal
  
- ▶ **7. Which term describes one of multiple logical consoles that can each support an independent login session?**
  - a. Physical Console
  - b. Virtual Console
  - c. Shell
  - d. Terminal
  
- ▶ **8. Which term describes an interface that provides a display for output and a keyboard for input to a shell session?**
  - a. Console
  - b. Virtual Console
  - c. Shell
  - d. Terminal

## ► Solution

# Accessing the Command Line

Choose the correct answer to the following questions:

- 1. Which term describes the interpreter that executes commands typed as strings?
  - a. Command
  - b. Console
  - c. Shell
  - d. Terminal
  
- 2. Which term describes the visual cue that indicates an interactive shell is waiting for the user to type a command?
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  - a. Console
  - b. Virtual Console
  - c. Shell
  - d. Terminal

# Configuring SSH Key-based Authentication

## Objectives

After completing this section, you should be able to configure a user account to use key-based authentication to log in to remote systems securely without a password.

## SSH Key-based Authentication

You can configure an SSH server to allow you to authenticate without a password by using key-based authentication. This is based on a private-public key scheme.

To do this, you generate a matched pair of cryptographic key files. One is a private key, the other a matching public key. The private key file is used as the authentication credential and, like a password, must be kept secret and secure. The public key is copied to systems the user wants to connect to, and is used to verify the private key. The public key does not need to be secret.

You put a copy of the public key in your account on the server. When you try to log in, the SSH server can use the public key to issue a challenge that can only be correctly answered by using the private key. As a result, your **ssh** client can automatically authenticate your login to the server with your unique copy of the private key. This allows you to securely access systems in a way that doesn't require you to enter a password interactively every time.

## Generating SSH Keys

To create a private key and matching public key for authentication, use the **ssh-keygen** command. By default, your private and public keys are saved in your **~/ .ssh/id\_rsa** and **~/ .ssh/id\_rsa.pub** files, respectively.

```
[user@host ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/user/.ssh/id_rsa): Enter
Created directory '/home/user/.ssh'.
Enter passphrase (empty for no passphrase): Enter
Enter same passphrase again: Enter
Your identification has been saved in /home/user/.ssh/id_rsa.
Your public key has been saved in /home/user/.ssh/id_rsa.pub.
The key fingerprint is:
SHA256:vxutUNPio3QDCyvkYm1oIx35hmMrHpPKWFdIYu3HV+w user@host.lab.example.com
The key's randomart image is:
+---[RSA 2048]---+
|
| .      .
| o o    o
| . = o   o .
| o + = S E .
| ..0 o + * +
| .+% 0 . + B .
|
```

```
|=*00 . . + * |
|++ . . +. |
+---- [SHA256] ----+
```

If you do not specify a passphrase when **ssh-keygen** prompts you, the generated private key is not protected. In this case, anyone with your private key file could use it for authentication. If you set a passphrase, then you will need to enter that passphrase when you use the private key for authentication. (Therefore, you would be using the private key's passphrase rather than your password on the remote host to authenticate.)

You can run a helper program called **ssh-agent** which can temporarily cache your private key passphrase in memory at the start of your session to get true passwordless authentication. This will be discussed later in this section.

The following example of the **ssh-keygen** command shows the creation of the passphrase-protected private key alongside the public key.

```
[user@host ~]$ ssh-keygen -f .ssh/key-with-pass
Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in .ssh/key-with-pass.
Your public key has been saved in .ssh/key-with-pass.pub.
The key fingerprint is:
SHA256:w3GGB7EyHUry4a0cNPKmhNKS7dl1YsMVLvFZJ77VxAo user@host.lab.example.com
The key's randomart image is:
+---[RSA 2048]----+
| . + =.0 ... |
| = B XEo o. |
| . o 0 X =... |
| = = B = o. |
| = + * * S . |
|. + = o + . |
| + . |
| |
| |
+---- [SHA256] ----+
```

The **-f** option with the **ssh-keygen** command determines the files where the keys are saved. In the preceding example, the private and public keys are saved in the **/home/user/.ssh/key-with-pass** **/home/user/.ssh/key-with-pass.pub** files, respectively.



### Warning

During further SSH keypair generation, unless you specify a unique file name, you are prompted for permission to overwrite the existing **id\_rsa** and **id\_rsa.pub** files. If you overwrite the existing **id\_rsa** and **id\_rsa.pub** files, then you must replace the old public key with the new one on all the SSH servers that have your old public key.

Once the SSH keys have been generated, they are stored by default in the **.ssh/** directory of the user's home directory. The permission modes must be 600 on the private key and 644 on the public key.

## Sharing the Public Key

Before key-based authentication can be used, the public key needs to be copied to the destination system. The **ssh-copy-id** command copies the public key of the SSH keypair to the destination system. If you omit the path to the public key file while running **ssh-copy-id**, it uses the default **/home/user/.ssh/id\_rsa.pub** file.

```
[user@host ~]$ ssh-copy-id -i .ssh/key-with-pass.pub user@remotehost
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/user/.ssh/
id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted
now it is to install the new keys
user@remotehost's password: redhat
Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'user@remotehost'"
and check to make sure that only the key(s) you wanted were added.
```

After the public key is successfully transferred to a remote system, you can authenticate to the remote system using the corresponding private key while logging in to the remote system over SSH. If you omit the path to the private key file while running the **ssh** command, it uses the default **/home/user/.ssh/id\_rsa** file.

```
[user@host ~]$ ssh -i .ssh/key-with-pass user@remotehost
Enter passphrase for key '.ssh/key-with-pass': redhatpass
...output omitted...
[user@remotehost ~]$ exit
logout
Connection to remotehost closed.
[user@host ~]$
```

## Using ssh-agent for Non-interactive Authentication

If your SSH private key is protected with a passphrase, you normally have to enter the passphrase to use the private key for authentication. However, you can use a program called **ssh-agent** to temporarily cache the passphrase in memory. Then any time that you use SSH to log in to another system with the private key, **ssh-agent** will automatically provide the passphrase for you. This is convenient, and can improve security by providing fewer opportunities for someone "shoulder surfing" to see you type the passphrase in.

Depending on your local system's configuration, if you initially log in to the GNOME graphical desktop environment, the **ssh-agent** program might automatically be started and configured for you.

If you log in on a text console, log in using **ssh**, or use **sudo** or **su**, you will probably need to start **ssh-agent** manually for that session. You can do this with the following command:

```
[user@host ~]$ eval $(ssh-agent)
Agent pid 10155
[user@host ~]$
```

**Note**

When you run **ssh-agent**, it prints out some shell commands. You need to run these commands to set environment variables used by programs like **ssh-add** to communicate with it. The **eval \$(ssh-agent)** command starts **ssh-agent** and runs those commands to automatically set those environment variables for that shell session. It also displays the PID of the **ssh-agent** process.

Once **ssh-agent** is running, you need to tell it the passphrase for your private key or keys. You can do this with the **ssh-add** command.

The following **ssh-add** commands add the private keys from **/home/user/.ssh/id\_rsa** (the default) and **/home/user/.ssh/key-with-pass** files, respectively.

```
[user@host ~]$ ssh-add
Identity added: /home/user/.ssh/id_rsa (user@host.lab.example.com)
[user@host ~]$ ssh-add .ssh/key-with-pass
Enter passphrase for .ssh/key-with-pass: redhatpass
Identity added: .ssh/key-with-pass (user@host.lab.example.com)
```

After successfully adding the private keys to the **ssh-agent** process, you can invoke an SSH connection using the **ssh** command. If you are using any private key file other than the default **/home/user/.ssh/id\_rsa** file, then you must use the **-i** option with the **ssh** command to specify the path to the private key file.

The following example of the **ssh** command uses the default private key file to authenticate to an SSH server.

```
[user@host ~]$ ssh user@remotehost
Last login: Fri Apr  5 10:53:50 2019 from host.example.com
[user@remotehost ~]$
```

The following example of the **ssh** command uses the **/home/user/.ssh/key-with-pass** (non-default) private key file to authenticate to an SSH server. The private key in the following example has already been decrypted and added to its parent **ssh-agent** process, so the **ssh** command does not prompt you to decrypt the private key by interactively entering its passphrase.

```
[user@host ~]$ ssh -i .ssh/key-with-pass user@remotehost
Last login: Mon Apr  8 09:44:20 2019 from host.example.com
[user@remotehost ~]$
```

When you log out of the session that started **ssh-agent**, the process will exit and your the passphrases for your private keys will be cleared from memory.

**References**

**ssh-keygen(1)**, **ssh-copy-id(1)**, **ssh-agent(1)**, **ssh-add(1)** man pages

## ► Guided Exercise

# Configuring SSH Key-based Authentication

In this exercise, you will configure a user to use key-based authentication for SSH.

## Outcomes

You should be able to:

- Generate an SSH key pair without passphrase protection.
- Generate an SSH key pair with passphrase protection.
- Authenticate using both passphrase-less and passphrase-protected SSH keys.

## Before You Begin

Log in to **workstation** as **student** using **student** as the password.

On **workstation**, run **lab ssh-configure start** to start the exercise. This script creates the necessary user accounts.

```
[student@workstation ~]$ lab ssh-configure start
```

- 1. From **workstation**, open an SSH session to **serverb** as **student**.

```
[student@workstation ~]$ ssh student@serverb
...output omitted...
[student@serverb ~]$
```

- 2. Use the **su** command to switch to the **operator1** user on **serverb**. Use **redhat** as the password of **operator1**.

```
[student@serverb ~]$ su - operator1
Password: redhat
[operator1@serverb ~]$
```

- 3. Use the **ssh-keygen** command to generate SSH keys. Do not enter a passphrase.

```
[operator1@serverb ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/operator1/.ssh/id_rsa): Enter
Created directory '/home/operator1/.ssh'.
Enter passphrase (empty for no passphrase): Enter
Enter same passphrase again: Enter
Your identification has been saved in /home/operator1/.ssh/id_rsa.
Your public key has been saved in /home/operator1/.ssh/id_rsa.pub.
```

```

The key fingerprint is:
SHA256:JainiQdnRosC+xxH0qsJQQLzBNULdb+jJbyrCZQBERI
operator1@serverb.lab.example.com
The key's randomart image is:
+---[RSA 2048]-----+
|E+*+ooo .          |
|. = 0.o 0 .          |
|o.. = . . 0          |
|+. + * . 0 .          |
|+ = X . S +          |
| + @ + = .          |
|. + = 0              |
|.0 . . . .          |
|o   o..              |
+----[SHA256]-----+

```

- 4. Use the **ssh-copy-id** command to send the public key of the SSH key pair to **operator1** on **servera**. Use **redhat** as the password of **operator1** on **servera**.

```

[operator1@serverb ~]$ ssh-copy-id operator1@servera
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/
operator1/.ssh/id_rsa.pub"
The authenticity of host 'servera (172.25.250.10)' can't be established.
ECDSA key fingerprint is SHA256:ERTdjoo0IrIwVSZQnqD5or+JbXfidg0udb3DXBuHWzA.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted
now it is to install the new keys
operator1@servera's password: redhat
Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'operator1@servera'"
and check to make sure that only the key(s) you wanted were added.

```

- 5. Execute the **hostname** command on **servera** remotely using SSH without accessing the remote interactive shell.

```

[operator1@serverb ~]$ ssh operator1@servera hostname
servera.lab.example.com

```

Notice that the preceding **ssh** command did not prompt you for a password because it used the passphrase-less private key against the exported public key to authenticate as **operator1** on **servera**. This approach is not secure, because anyone who has access to the private key file can log in to **servera** as **operator1**. The secure alternative is to protect the private key with a passphrase, which is the next step.

- 6. Use the **ssh-keygen** command to generate another set of SSH keys with passphrase-protection. Save the key as **/home/operator1/.ssh/key2**. Use **redhatpass** as the passphrase of the private key.

**Warning**

If you do not specify the file where the key gets saved, the default file (**/home/user/.ssh/id\_rsa**) is used. You have already used the default file name when generating SSH keys in the preceding step, so it is vital that you specify a non-default file, otherwise the existing SSH keys will be overwritten.

```
[operator1@serverb ~]$ ssh-keygen -f .ssh/key2
Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase): redhatpass
Enter same passphrase again: redhatpass
Your identification has been saved in .ssh/key2.
Your public key has been saved in .ssh/key2.pub.
The key fingerprint is:
SHA256:0CtCjFPm5QrbPBgbEIWCcw5AI4oSLMEbgLrBQ1HWKI
operator1@serverb.lab.example.com
The key's randomart image is:
+---[RSA 2048]-----+
|O=X*                |
|OB=.                |
|E*o.               |
|Booo .             |
|..= . o S          |
|+.o  o             |
|+.oo+ o            |
|+o.O.+             |
|+ . =o.            |
+---[SHA256]-----+
```

- 7. Use the **ssh-copy-id** command to send the public key of the passphrase-protected key pair to **operator1** on **servera**.

```
[operator1@serverb ~]$ ssh-copy-id -i .ssh/key2.pub operator1@servera
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ".ssh/key2.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted
now it is to install the new keys
```

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'operator1@servera'"  
and check to make sure that only the key(s) you wanted were added.

Notice that the preceding **ssh-copy-id** command did not prompt you for a password because it used the public key of the passphrase-less private key that you exported to **servera** in the preceding step.

- ▶ 8. Execute the **hostname** command on **servera** remotely with SSH without accessing the remote interactive shell. Use **/home/operator1/.ssh/key2** as the identity file. Specify **redhatpass** as the passphrase, which you set for the private key in the preceding step.

```
[operator1@serverb ~]$ ssh -i .ssh/key2 operator1@servera hostname
Enter passphrase for key '.ssh/key2': redhatpass
servera.lab.example.com
```

Notice that the preceding **ssh** command prompted you for the passphrase you used to protect the private key of the SSH key pair. This passphrase protects the private key. Should an attacker gain access to the private key, the attacker cannot use it to access other systems because the private key itself is protected with a passphrase. The **ssh** command uses a different passphrase than the one for **operator1** on **servera**, requiring users to know both.

You can use **ssh-agent**, as in the following step, to avoid interactively typing in the passphrase while logging in with SSH. Using **ssh-agent** is both more convenient and more secure in situations where the administrators log in to remote systems regularly.

- ▶ 9. Run **ssh-agent** in your Bash shell and add the passphrase-protected private key (**/home/operator1/.ssh/key2**) of the SSH key pair to the shell session.

```
[operator1@serverb ~]$ eval $(ssh-agent)
Agent pid 21032
[operator1@serverb ~]$ ssh-add .ssh/key2
Enter passphrase for .ssh/key2: redhatpass
Identity added: .ssh/key2 (operator1@serverb.lab.example.com)
```

The preceding **eval** command started **ssh-agent** and configured this shell session to use it. You then used **ssh-add** to provide the unlocked private key to **ssh-agent**.

- ▶ 10. Execute the **hostname** command on **servera** remotely without accessing a remote interactive shell. Use **/home/operator1/.ssh/key2** as the identity file.

```
[operator1@serverb ~]$ ssh -i .ssh/key2 operator1@servera hostname
servera.lab.example.com
```

Notice that the preceding **ssh** command did not prompt you to enter the passphrase interactively.

- ▶ 11. Open another terminal on **workstation** and open an SSH session to **serverb** as **student**.

```
[student@workstation ~]$ ssh student@serverb
...output omitted...
[student@serverb ~]$
```

- ▶ 12. On **serverb**, use the **su** command to switch to **operator1** and invoke an SSH connection to **servera**. Use **/home/operator1/.ssh/key2** as the identity file to authenticate using the SSH keys.

12.1. Use the **su** command to switch to **operator1**. Use **redhat** as the password of **operator1**.

```
[student@serverb ~]$ su - operator1
Password: redhat
[operator1@serverb ~]$
```

12.2. Open an SSH session to **servera** as **operator1**.

```
[operator1@serverb ~]$ ssh -i .ssh/key2 operator1@servera
Enter passphrase for key '.ssh/key2': redhatpass
...output omitted...
[operator1@servera ~]$
```

Notice that the preceding **ssh** command prompted you to enter the passphrase interactively because you did not invoke the SSH connection from the shell that you used to start **ssh-agent**.

► 13. Exit all the shells you are using in the second terminal.

13.1. Log out of **servera**.

```
[operator1@servera ~]$ exit
logout
Connection to servera closed.
[operator1@serverb ~]$
```

13.2. Exit the **operator1** and **student** shells on **serverb** to return to the **student** user's shell on **workstation**.

```
[operator1@serverb ~]$ exit
logout
[student@serverb ~]$ exit
logout
Connection to serverb closed.
[student@workstation ~]$
```

13.3. Close the second terminal on **workstation**.

```
[student@workstation ~]$ exit
```

► 14. Log out of **serverb** on the first terminal and conclude this exercise.

14.1. From the first terminal, exit the **operator1** user's shell on **serverb**.

```
[operator1@serverb ~]$ exit
logout
[student@serverb ~]$
```

The **exit** command caused you to exit the **operator1** user's shell, terminating the shell session where **ssh-agent** was active, and return to the **student** user's shell on **serverb**.

14.2. Exit the **student** user's shell on **serverb** to return to the **student** user's shell on **workstation**.

```
[student@serverb ~]$ exit  
logout  
Connection to serverb closed.  
[student@workstation ~]$
```

## Finish

On **workstation**, run **lab ssh-configure finish** to complete this exercise.

```
[student@workstation ~]$ lab ssh-configure finish
```

This concludes the guided exercise.

# Getting Help From Red Hat Customer Portal

## Objectives

After completing this section, you should be able to describe key resources available through the Red Hat Customer Portal and use them to find information from Red Hat documentation and the Knowledgebase.

## Accessing Support Resources on the Red Hat Customer Portal

The Red Hat Customer Portal (<https://access.redhat.com>) provides customers access to documentation, downloads, tools, and technical expertise. Customers can search for solutions, FAQs, and articles through the Knowledgebase. From the Customer Portal, you can:

- Access official product documentation.
- Submit and manage support tickets.
- Manage software subscriptions and entitlements.
- Obtain software downloads, updates, and evaluations.
- Consult tools that can help you optimize the configuration of your systems.

Parts of the site are accessible to everyone, and other areas are only available to customers with active subscriptions. Get help accessing the Customer Portal at <https://access.redhat.com/help/>.

## Getting Oriented to the Customer Portal

You can access the Red Hat Customer Portal through a web browser. This section introduces the Customer Portal Tour. The tour can be found at <https://access.redhat.com/start>.

The tour is a very useful tool for discovering all the portal has to offer and how to get the most out of your Red Hat subscription. After you have logged in to the Red Hat Customer Portal, click **Tour the Customer Portal**.

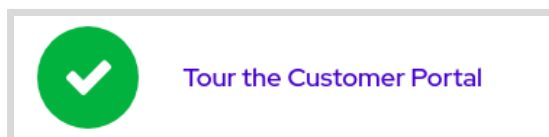


Figure 1.1: Tour the Customer Portal

The **WELCOME TO THE RED HAT CUSTOMER PORTAL** window opens with two options: **CLOSE** and **NEXT**. Click **NEXT** to start the tour. This is the first of a sequence of windows that highlight different parts of the interface.

## The Top Navigation Bar

The first three steps on the Customer Portal Tour can be found on the top navigation bar of the Red Hat Customer Portal website:



Figure 1.2: Top Navigation Bar

**Subscriptions** opens a new page where you can manage your registered systems and your subscriptions and entitlements usage. It lists information about errata that apply and allows you to create *activation keys* that you can use when registering systems to ensure they get entitlements from the correct subscriptions. Note that if you are part of an organization, your Organization Administrator can limit your access to this page.

**Downloads** opens a new page which gives you access to your product downloads and to request evaluation entitlements for products for which you do not have entitlements.

**Support Cases** opens a new page which provides access to create, track, and manage your support cases through the Case Management system, assuming that your organization has authorized that level of access.

Your name is the title for the **User Menu**, which allows you to manage your account, accounts for which you are Organization Administrator, your personal profile, and options for email notifications of new content that is available.

The globe icon opens the **Select Your Language** menu to specify your language preferences for Customer Portal.

## Topics Menus

Underneath the top navigation bar on the Customer Portal's main page are menus that you can use to navigate to four major categories of resources available on the site.

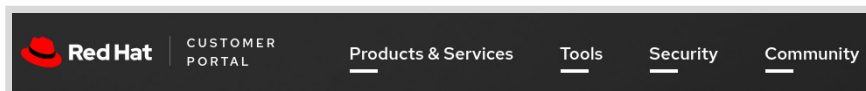


Figure 1.3: Resources Menus

**Products & Services** provides access to the *Product Hubs*, pages that provide access to product-specific evaluations, overviews, getting started guides, and other product support information. You can also access documentation for Red Hat products, direct links to the Knowledgebase of support articles, and information on support policies and how to contact Red Hat Support.

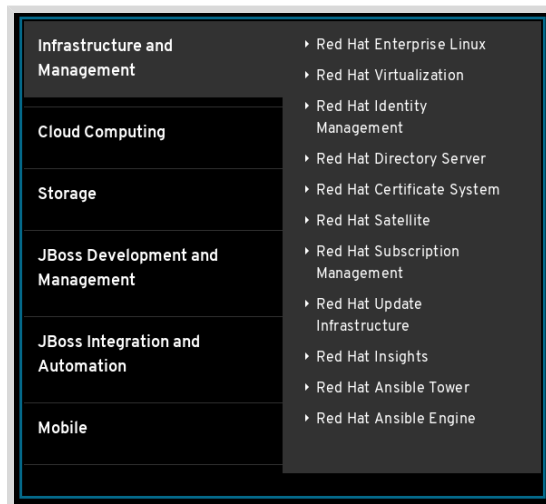


Figure 1.4: Products and Services

The **Tools** menu provides links to tools to help you succeed with Red Hat products. The Solution Engine section provides you with an efficient way to search for solutions to your problems quickly, by product, and opening a support ticket if you do not find a satisfactory solution. The Customer Portal Labs section provides a collection of web-based applications and tools to help you improve performance, diagnose issues, identify security problems, and optimize your configurations. For example, the Product Life Cycle Checker allows you to select a particular product and view its support life cycle schedule. Another tool, the Rescue Mode Assistant, helps you reset the root password of a system, generate diagnostic reports, or fix boot-time problems with file systems. But there are many other tools available at that site.

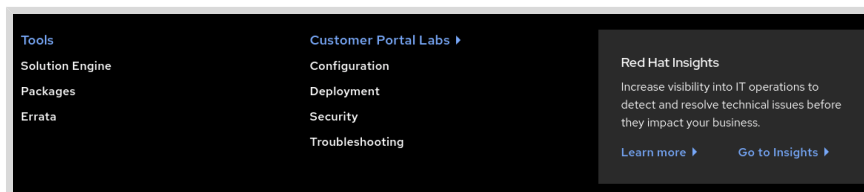


Figure 1.5: Tools menu in Customer Portal

The Security section provides access to the *Red Hat Product Security Center* at <https://access.redhat.com/security/>. This section also provides information about high-profile security issues, access to the Red Hat CVE Database, the Security channel of the Red Hat Blog, and resources about Red Hat's security response process and how we rate issues and resolve them.

Finally, the Community section is a place where Red Hat experts, customers, and partners can communicate and collaborate. Discussion forums, blogs, and information about upcoming events in your area are available here.

**Note**

You should complete the entire tour at Getting Started with Red Hat [<https://access.redhat.com/start>], including the sections on how to personalize your Customer Portal experience and exploring the benefits of your Red Hat subscription, to get the full story about the Customer Portal. You will need at least one active subscription on your Customer Portal account to access this page.

## Searching the Knowledgebase with the Red Hat Support Tool

The Red Hat Support Tool utility, **redhat-support-tool**, provides a text-based interface that allows you to search Knowledgebase articles and to file support cases on the Customer Portal from your system's command line. The tool does not have a graphical interface and, because it interacts with the Red Hat Customer Portal, it requires internet access. Run the **redhat-support-tool** command using any terminal or SSH connection.

The **redhat-support-tool** command may be used in an interactive mode or invoked as a command with options and arguments. The tool's syntax is identical for both methods. By default, the program launches in interactive mode. Use the **help** subcommand to see all available commands. Interactive mode supports tab completion and the ability to call programs in the parent shell.

```
[user@host ~]$ redhat-support-tool
Welcome to the Red Hat Support Tool.
Command (? for help):
```

When first invoked, **redhat-support-tool** prompts for Red Hat Customer Portal subscriber login information. To avoid repetitively supplying this information, the tool asks to store account information in the user's home directory (`~/redhat-support-tool/redhat-support-tool.conf`). If issues are all filed through a particular Red Hat Customer Portal account, the **--global** option can save account information to `/etc/redhat-support-tool.conf`, along with other system-wide configuration. The tool's **config** command modifies tool configuration settings.

The **redhat-support-tool** command allows subscribers to search and display Knowledgebase content from the Red Hat Customer Portal. The Knowledgebase permits keyword searches, similar to the **man** command. You can enter error codes, syntax from log files, or any mix of keywords to produce a list of relevant solution documents.

The following is an initial configuration and basic search demonstration:

```
[user@host ~]$ redhat-support-tool
Welcome to the Red Hat Support Tool.
Command (? for help): search How to manage system entitlements with subscription-
manager
Please enter your RHN user ID: subscriber
Save the user ID in /home/student/.redhat-support-tool/redhat-support-tool.conf
(y/n): y
Please enter the password for subscriber: password
Save the password for subscriber in /home/student/.redhat-support-tool/redhat-
support-tool.conf (y/n): y
```

After prompting the user for the required user configuration, the tool continues with the original search request:

```
Type the number of the solution to view or 'e' to return to the previous menu.
1 [ 253273:VER] How to register and subscribe a system to the Red Hat Customer
Portal using Red Hat Subscription-Manager
2 [ 265523:VER] Enabling or disabling a repository using Red Hat Subscription
Management
```

```

3 [ 100423:VER] Why does subscription-manager list return: "No Installed
Products found" ?
...output omitted...
Select a Solution: 1

```

Select article number 1 as above and you are prompted to select the section of the document to read. Finally, use the **Q** key to quit the section you are in, or use it repeatedly to quit the **redhat-support-tool** command.

```

Select a Solution: 1

Type the number of the section to view or 'e' to return to the previous menu.
1 Title
2 Issue
3 Environment
4 Resolution
5 Display all sections
End of options.
Section: 1

Title
=====
How to register and subscribe a system to the Red Hat Customer Portal using Red
Hat Subscription-Manager
URL:      https://access.redhat.com/solutions/253273
Created On:  None
Modified On: 2017-11-29T15:33:51Z

(END) q
Section:
Section: q

Select a Solution: q

Command (? for help): q
[user@hosts ~]#

```

## Accessing Knowledgebase Articles by Document ID

Locate online articles directly using the tool's **kb** command with the Knowledgebase document ID. A returned document scrolls on the screen without pagination, but you can redirect it to a file to save it and use **less** to scroll through it a screen at a time.

```

[user@host ~]$ redhat-support-tool kb 253273

Title
=====
How to register and subscribe a system to the Red Hat Customer Portal using Red
Hat Subscription-Manager
URL:      https://access.redhat.com/solutions/253273
Created On:  None
Modified On: 2017-11-29T15:33:51Z

```

## Issue

```
=====
*   How to register a new `Red Hat Enterprise Linux` system to the Customer Portal
    using `Red Hat Subscription-Manager`
...output omitted...
```

## Managing Support Cases with Red Hat Support Tool

One benefit of a product subscription is access to technical support through the Red Hat Customer Portal. Depending on the system's subscription support level, Red Hat may be contacted through online tools or by phone. See [https://access.redhat.com/site/support/policy/support\\_process](https://access.redhat.com/site/support/policy/support_process) for detailed information.

### Preparing a Bug report

Before contacting Red Hat Support, it is important to gather relevant information for a bug report.

*Define the problem.* Be able to clearly state the problem and its symptoms. Be as specific as possible. Detail the steps that will reproduce the problem.

*Gather background information.* Which product and version is affected? Be ready to provide relevant diagnostic information. This can include output of **sosreport**, discussed later in this section. For kernel problems, this could include the system's **kdump** crash dump or a digital photo of the kernel backtrace displayed on the monitor of a crashed system.

*Determine the severity level.* Red Hat uses four severity levels to classify issues. *Urgent* and *High* severity problem reports should be followed by a phone call to the relevant local support center (see <https://access.redhat.com/site/support/contact/technicalSupport>).

| Severity                   | Description                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Urgent</i> (Severity 1) | A problem that severely impacts your use of the software in a production environment. This includes loss of production data or malfunctioning production systems. The situation halts your business operations and no procedural workaround exists.                                                                                                                                                                  |
| <i>High</i> (Severity 2)   | A problem where the software is functioning but use in a production environment is severely reduced. The situation is causing a high impact to your business operations and no procedural workaround exists.                                                                                                                                                                                                         |
| <i>Medium</i> (Severity 3) | A problem that involves partial, non critical loss of use of the software in a production environment or development environment. For production environments, there is a medium to low impact on your business. Business continues to function using a procedural workaround. For development environments, the situation is causing problems migrating your project into production.                               |
| <i>Low</i> (Severity 4)    | A general usage question, reporting of a documentation error, or recommendation for a future product enhancement or modification. For production environments, there is low to no impact on your business or the performance or functionality of your system. For development environments, there is a medium to low impact on your business, but your business continues to function using a procedural workaround. |

## Managing a Bug Report with `redhat-support-tool`

You can create, view, modify, and close Red Hat Support cases using `redhat-support-tool`. When support cases are in an **opened** or **maintained** status, users may attach files or documentation, such as diagnostic reports (sosreport). The tool uploads and attaches files to cases.

Case details including the product name, version, summary, description, severity, and case group may be assigned with command options or letting the tool prompt for required information. In the following example, a new case is opened. The `--product` and `--version` options are specified.

```
[user@host ~]$ redhat-support-tool
Welcome to the Red Hat Support Tool.
Command (? for help): opencase --product="Red Hat Enterprise Linux" --
version="7.0"
Please enter a summary (or 'q' to exit): System fails to run without power
Please enter a description (Ctrl-D on an empty line when complete):
When the server is unplugged, the operating system fails to continue.
 1 Urgent
 2 High
 3 Normal
 4 Low
Please select a severity (or 'q' to exit): 4
Would you like to assign a case group to this case (y/N)? N
Would see if there is a solution to this problem before opening a support case?
(y/N) N
-----
Support case 01034421 has successfully been opened.
```

If the `--product` and `--version` options are not specified the `redhat-support-tool` provides a list of choices for those options.

```
[user@host ~]$ redhat-support-tool
Welcome to the Red Hat Support Tool.
Command (? for help): opencase
Do you want to use the default product - "Red Hat Enterprise Linux" (y/N)? y
...output omitted...
29 7.4
30 7.5
31 7.6
32 8.0 Beta
Please select a version (or 'q' to exit): 32
Please enter a summary (or 'q' to exit): yum fails to install apache
Please enter a description (Ctrl-D on an empty line when complete):
yum cannot find correct repo
 1 Urgent
 2 High
 3 Normal
 4 Low
Please select a severity (or 'q' to exit): 4
Would you like to use the default (Ungrouped Case) Case Group (y/N)? : y
```

```
Would you like to see if there's a solution to this problem before opening a
support case? (y/N) N
```

```
-----
Support case 010355678 has successfully been opened.
```

## Attaching Diagnostic Information to a Support Case

Including diagnostic information can lead to a quicker resolution. Attach the *sosreport* when the case is opened. The **sosreport** command generates a compressed tar archive of diagnostic information gathered from the running system. The **redhat-support-tool** prompts to include one if an archive has been created previously:

```
Please attach a SoS report to support case 01034421. Create a SoS report as
the root user and execute the following command to attach the SoS report
directly to the case:
```

```
redhat-support-tool addattachment -c 01034421 path to sosreport
```

```
Would you like to attach a file to 01034421 at this time? (y/N) N
Command (? for help):
```

If a current SoS report does not exist, an administrator can generate and attach one later. Use the **redhat-support-tool addattachment** command to attach the report.

Support cases can also be viewed, modified, and closed by the subscriber:

```
Command (? for help): listcases
```

```
Type the number of the case to view or 'e' to return to the previous menu.
```

```
1 [Waiting on Red Hat] System fails to run without power
```

```
No more cases to display
```

```
Select a Case: 1
```

```
Type the number of the section to view or 'e' to return to the previous menu.
```

```
1 Case Details
```

```
2 Modify Case
```

```
3 Description
```

```
4 Recommendations
```

```
5 Get Attachment
```

```
6 Add Attachment
```

```
7 Add Comment
```

```
End of options.
```

```
Option: q
```

```
Select a Case: q
```

```
Command (? for help):q
```

```
[user@host ~]$ redhat-support-tool modifycase --status=Closed 01034421
```

```
Successfully updated case 01034421
```

```
[user@host ~]$
```

The Red Hat Support Tool has advanced application diagnostic and analytic capabilities. Using kernel crash dump core files, **redhat-support-tool** can create and extract a *backtrace*. A

backtrace is a report of the active stack frames at the point of a crash dump and provides onsite diagnostics. One of the options of the **redhat-support-tool** is to open a support case.

The tool also provides log file analysis. Using the tool's **analyze** command, log files of many types, including operating system, JBoss, Python, Tomcat, and oVirt, can be parsed to recognize problem symptoms. The log files can be viewed and diagnosed individually. Providing preprocessed analysis, as opposed to raw data such as crash dump or log files, allows support cases to be opened and made available to engineers more quickly.

## Joining Red Hat Developer

One other useful resource available from Red Hat is Red Hat Developer. Hosted at <https://developer.redhat.com>, this program provides subscription entitlements to Red Hat software for development purposes, documentation, and premium books from our experts on microservices, serverless computing, Kubernetes, and Linux. A blog, links to information about upcoming events and training, and other help resources are also available, as well as links to Red Hat Customer Portal.

Registration is free, and can be completed at <https://developer.redhat.com/register>.



### References

**sosreport**(1) man page

#### Red Hat Access: Red Hat Support Tool

<https://access.redhat.com/site/articles/445443>

#### Red Hat Support Tool First Use

<https://access.redhat.com/site/videos/534293>

#### Contacting Red Hat Technical Support

[https://access.redhat.com/site/support/policy/support\\_process/](https://access.redhat.com/site/support/policy/support_process/)

#### Help - Red Hat Customer Portal

<https://access.redhat.com/site/help/>

## ► Guided Exercise

# Getting Help from Red Hat Customer Portal

In this exercise, you will generate a diagnostics report using Web Console.

## Outcomes

You should be able to generate a diagnostics report using Web Console which could be submitted to Red Hat Customer Portal as part of a support case.

## Before You Begin

Log in as the **student** user on **workstation** using **student** as the password.

From **workstation**, run the **lab support-portal start** command. The command runs a start script that determines if **servera** is reachable on the network. It also starts and enables Web Console on **servera**.

```
[student@workstation ~]$ lab support-portal start
```

- 1. From **workstation** use the **ssh** command to log into **servera** as the **student** user.

```
[student@workstation ~]$ ssh student@servera
Web console: https://servera.lab.example.com:9090/ or https://172.25.250.10:9090/
[student@servera ~]$
```

- 2. Use the **systemctl** command to confirm that the **cockpit** service is running. Enter **student** as the password when prompted.

```
[student@servera ~]$ systemctl status cockpit.socket
● cockpit.socket - Cockpit Web Service Socket
   Loaded: loaded (/usr/lib/systemd/system/cockpit.socket; enabled; vendor preset: disabled)
   Active: active (listening) since Thu 2019-05-16 10:32:33 IST; 4min 37s ago
     Docs: man:cockpit-ws(8)
    Listen: [::]:9090 (Stream)
   Process: 676 ExecStartPost=/bin/ln -snf active.motd /run/cockpit/motd (code=exited, status=0/SUCCESS)
   Process: 668 ExecStartPost=/usr/share/cockpit/motd/update-motd localhost (code=exited, status=0/SUCCESS)
      Tasks: 0 (limit: 11405)
     Memory: 1.5M
    CGroup: /system.slice/cockpit.socket
...output omitted...
```

- 3. Log out from **servera**.

```
[student@servera ~]$ exit  
[student@workstation ~]$
```

- ▶ **4.** On **workstation**, open *Firefox* and log in to the Web Console interface running on **servera.lab.example.com** as the **root** user with **redhat** as the password.
  - 4.1. Open *Firefox* and go to the **https://servera.lab.example.com:9090** address.
  - 4.2. If prompted, accept the self-signed certificate by adding it as an exception.
  - 4.3. Log in as the **root** user with **redhat** as the password. You are now logged in as a privileged user, which is necessary to create a diagnostic report.
  - 4.4. Click **Diagnostic Reports** in the left navigation bar. Click on **Create Report**. The report takes a few minutes to create.
- ▶ **5.** When the report is ready, click on **Download report**. Save the file.
  - 5.1. Click the **Download report** button, followed by the **Save File** button.
  - 5.2. Click the **Close** button.
  - 5.3. Log out from the Web Console interface.

## Finish

On **workstation**, run the **lab support-portal finish** script to complete this exercise.

```
[student@workstation ~]$ lab support-portal finish
```

This concludes the guided exercise.

# Detecting and Resolving Issues with Red Hat Insights

## Objectives

After completing this section, you should be able to use Red Hat Insights to analyze servers for issues, remediate or resolve them, and confirm the solution worked.

## Introducing Red Hat Insights

Red Hat Insights is a predictive analytics tool to help you identify and remediate threats to security, performance, availability, and stability on systems in your infrastructure running Red Hat products. Insights is delivered as a Software-as-a-Service (SaaS) product, so that you can deploy and scale it quickly with no additional infrastructure requirements. In addition, you can immediately take advantage of the latest recommendations and updates from Red Hat specific to your deployed systems.

Red Hat regularly updates the knowledge base used by Insights, based on common support risks, security vulnerabilities, known-bad configurations, and other issues identified by Red Hat. Actions to mitigate or remediate these issues are validated and verified by Red Hat. This support allows you to proactively identify, prioritize, and resolve issues before they become a larger problem.

For each detected issue, Insights provides estimates of the risk presented and recommendations on how to mitigate or remediate the problem. These recommendations may provide materials such as Ansible Playbooks or human-readable step-by-step instructions to help you resolve the issue.

Insights tailors recommendations to each system registered to the service. You install each client system with an agent that collects metadata about the runtime configuration of the system. This data is a subset of what you might provide to Red Hat Support using the **sosreport** command in order to resolve a support ticket. You can limit or obfuscate the data that your clients send. This blocks some of the analytic rules from operating, depending on what you limit.

Almost immediately after you register a server and it completes the initial system metadata synchronization, you should be able to see your server and any recommendations for it in the Insights console in Red Hat Cloud Portal.

Insights currently provides predictive analytics and recommendations for these Red Hat products:

- Red Hat Enterprise Linux 6.4 and later
- Red Hat Virtualization 4 and later
- Red Hat OpenShift Container Platform
- Red Hat OpenStack Platform 7 and later

## Describing the Insights Architecture

When you register a system with Insights, it immediately sends metadata about its current configuration to the Insights platform. After registration, the system periodically updates the metadata provided to Insights. The system sends the metadata using TLS encryption to protect it in transit.

When Insights receives the data, it analyses it and displays the result on the Insights web console at <https://cloud.redhat.com/insights>.

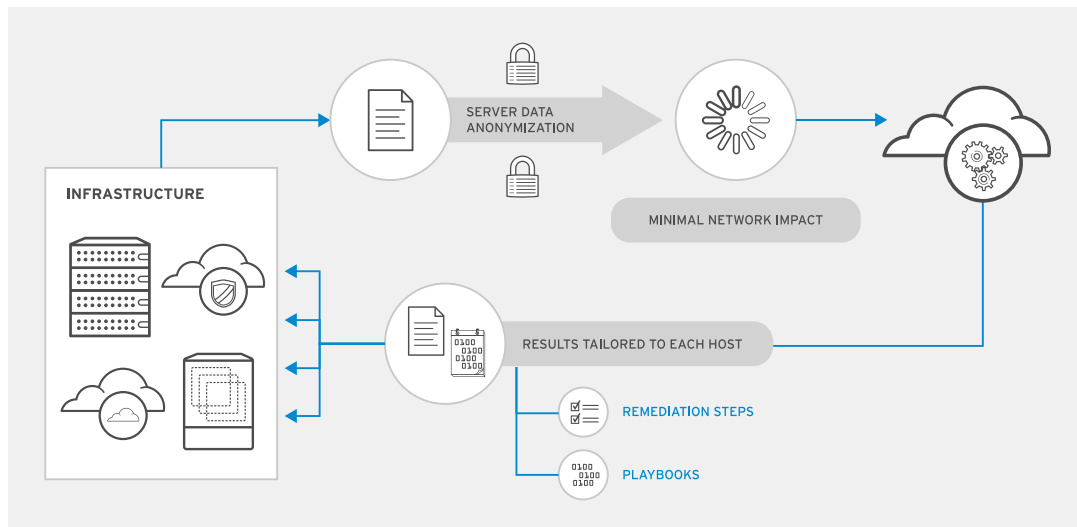


Figure 1.6: Insights high-level architecture

## Installing Insights Clients

Insights is included with Red Hat Enterprise Linux 8 as part of the subscription. Earlier versions of Red Hat Enterprise Linux servers require installing the *insights-client* package on the system.



### Important

The *insights-client* package replaces the *redhat-access-insights* package starting with Red Hat Enterprise Linux 7.5.

If your system is registered for software entitlements through the Customer Portal Subscription Management service, you can activate Insights with one command. Use the **insights-client --register** command to register the system.

```
[root@host ~]# insights-client --register
```

The Insights client periodically updates the metadata provided to Insights. Use the **insights-client** command to refresh the client's metadata at any time.

```
[root@host ~]# insights-client
Starting to collect Insights data for host.example.com
Uploading Insights data.
Successfully uploaded report for host.example.com.
View details about this system on cloud.redhat.com:
https://cloud.redhat.com/insights/inventory/dc480efd-4782-417e-a496-cb33e23642f0
```

## Registering a RHEL System with Insights

To register a RHEL server to Insights, the process is as follows:

- Interactively register the system with the Red Hat Subscription Management service.

```
[root@host ~]# subscription-manager register --auto-attach
```

- Make sure that the *insights-client* package is installed on the system. In RHEL 7, this package is in the **rhel-7-server-rpms** channel.

**Note**

This step is not required on Red Hat Enterprise Linux 8 systems.

```
[root@host ~]# yum install insights-client
```

- Use the **insights-client --register** command to register the system with the Insights service and upload initial system metadata.

```
[root@host ~]# insights-client --register
```

- Confirm that the system is visible under **Inventory** in the Insights web console at <https://cloud.redhat.com/insights>.

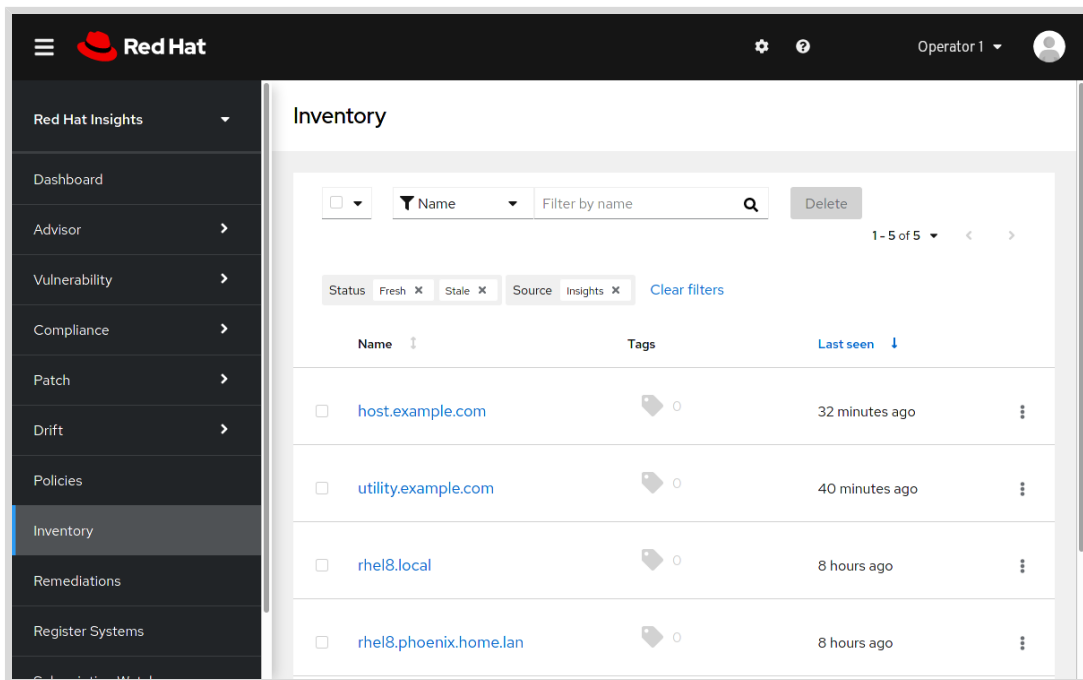


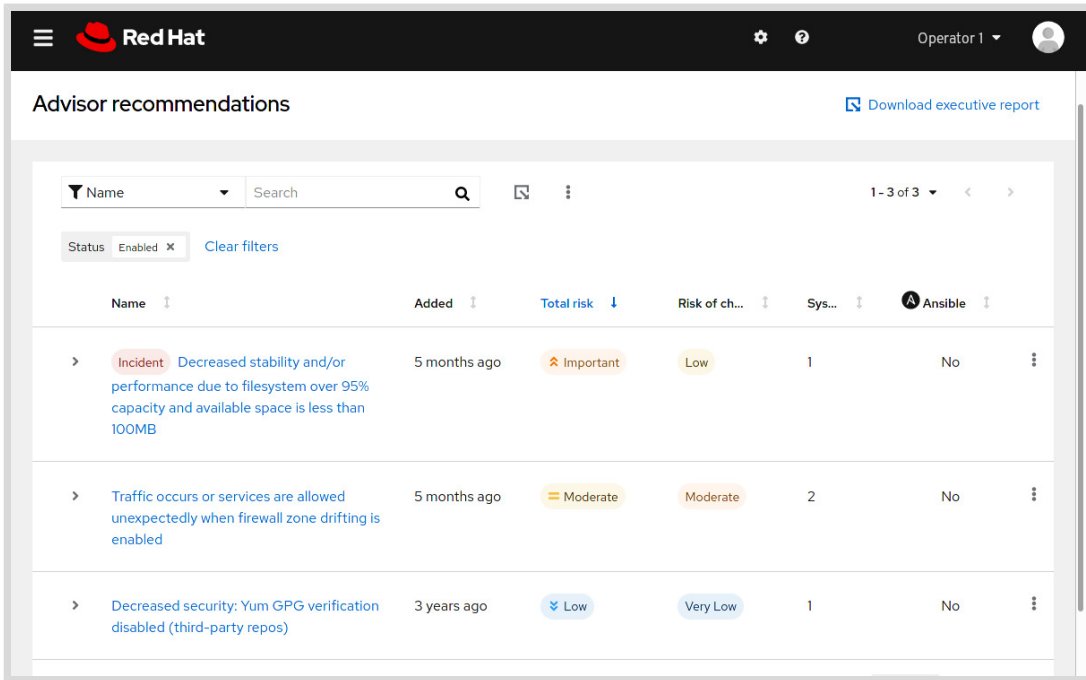
Figure 1.7: Insights Inventory on the Cloud Portal

## Navigating the Insights Console

Insights provides a family of services that you access through the web console at <https://cloud.redhat.com/insights>.

## Detecting Configuration Issues Using the Advisor Service

The Advisor service reports configuration issues that impact your systems. You access the service from the **Advisor** → **Recommendations** menu.

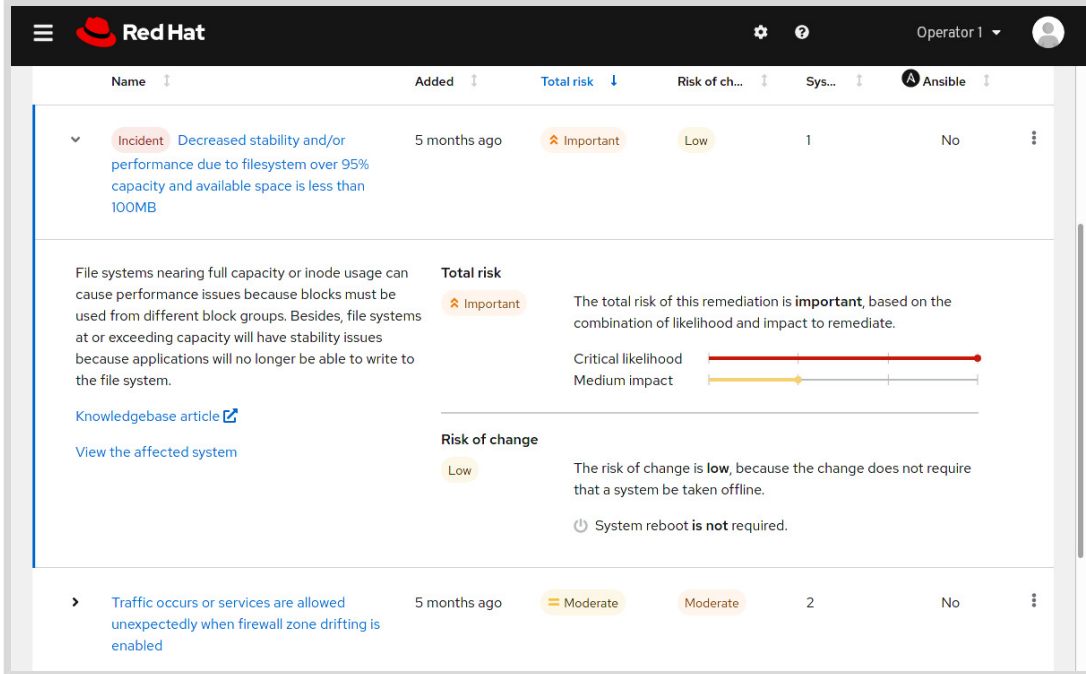


The screenshot shows the Red Hat Advisor recommendations page. At the top, there's a header with the Red Hat logo and a user profile. Below the header, the title 'Advisor recommendations' is displayed. A search bar and a 'Download executive report' link are also present. The main content area shows a table of recommendations with columns for Name, Added, Total risk, Risk of change, Sys..., and Ansible. Three recommendations are listed:

| Name                                                                                                                        | Added        | Total risk | Risk of change | Sys... | Ansible |
|-----------------------------------------------------------------------------------------------------------------------------|--------------|------------|----------------|--------|---------|
| Incident: Decreased stability and/or performance due to filesystem over 95% capacity and available space is less than 100MB | 5 months ago | Important  | Low            | 1      | No      |
| Traffic occurs or services are allowed unexpectedly when firewall zone drifting is enabled                                  | 5 months ago | Moderate   | Moderate       | 2      | No      |
| Decreased security: Yum GPG verification disabled (third-party repos)                                                       | 3 years ago  | Low        | Very Low       | 1      | No      |

Figure 1.8: Recommendations from the Advisor Service

For each issue, Insights provides additional information to help you understand the problem, prioritize work to address it, determine what mitigation or remediation is available, and automate resolution with an Ansible Playbook. Insights also provides links to Knowledgebase articles on the Customer Portal.



The screenshot shows the details for the first issue from Figure 1.8. The table lists the issue name, added date, total risk, risk of change, and system count. Below the table, there's a detailed description of the issue, a link to the Knowledgebase article, and a link to view the affected system. The 'Total risk' section explains that the risk is important based on likelihood and impact. The 'Risk of change' section explains that the risk is low because the change does not require a system reboot.

| Name                                                                                                                        | Added        | Total risk | Risk of change | Sys... | Ansible |
|-----------------------------------------------------------------------------------------------------------------------------|--------------|------------|----------------|--------|---------|
| Incident: Decreased stability and/or performance due to filesystem over 95% capacity and available space is less than 100MB | 5 months ago | Important  | Low            | 1      | No      |

**Total risk**  
Important  
The total risk of this remediation is **important**, based on the combination of likelihood and impact to remediate.

**Risk of change**  
Low  
The risk of change is **low**, because the change does not require that a system be taken offline.  
System reboot is **not** required.

Figure 1.9: Details of an Issue

The Advisor service evaluates the risk that an issue presents to your system in two categories.

### Total risk

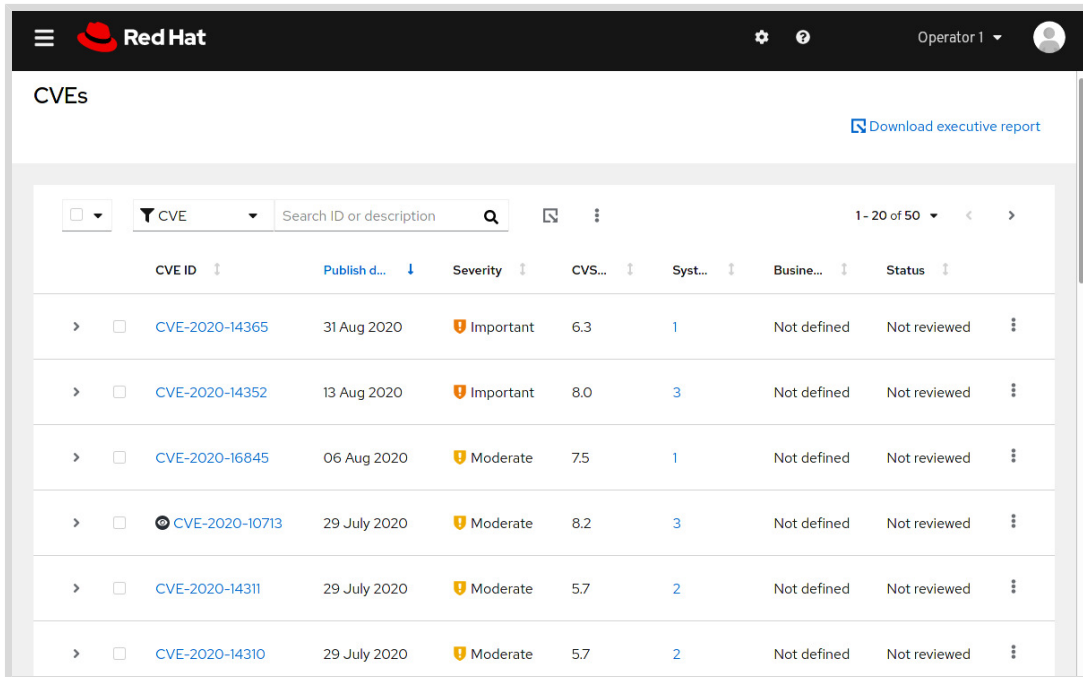
Indicates the impact of the issue on your system.

### Risk of change

Indicates the impact of the remediation action to your system. For example, you may have to restart the system.

## Assessing Security Using the Vulnerability Service

The Vulnerability service reports *Common Vulnerabilities and Exposures (CVEs)* that impact your systems. You access the service from the **Vulnerability** → **CVEs** menu.



The screenshot shows the Red Hat Vulnerability Service interface. At the top, there's a header with the Red Hat logo and a user profile labeled 'Operator 1'. Below the header, the page is titled 'CVEs' with a 'Download executive report' link. A search bar and a filter dropdown are present. The main content is a table of CVEs with columns for CVE ID, Publish date, Severity, CVSS score, Systems affected, Business impact, and Status. The table lists several CVEs, including CVE-2020-14365, CVE-2020-14352, CVE-2020-16845, CVE-2020-10713, CVE-2020-14311, and CVE-2020-14310.

| CVE ID         | Publish d... | Severity  | CVS... | Syst... | Busine...   | Status       |
|----------------|--------------|-----------|--------|---------|-------------|--------------|
| CVE-2020-14365 | 31 Aug 2020  | Important | 6.3    | 1       | Not defined | Not reviewed |
| CVE-2020-14352 | 13 Aug 2020  | Important | 8.0    | 3       | Not defined | Not reviewed |
| CVE-2020-16845 | 06 Aug 2020  | Moderate  | 7.5    | 1       | Not defined | Not reviewed |
| CVE-2020-10713 | 29 July 2020 | Moderate  | 8.2    | 3       | Not defined | Not reviewed |
| CVE-2020-14311 | 29 July 2020 | Moderate  | 5.7    | 2       | Not defined | Not reviewed |
| CVE-2020-14310 | 29 July 2020 | Moderate  | 5.7    | 2       | Not defined | Not reviewed |

Figure 1.10: Report from the Vulnerability Service

For each CVE, Insights provides additional information and lists the exposed systems. You can click **Remediate** to create an Ansible Playbook for remediation.

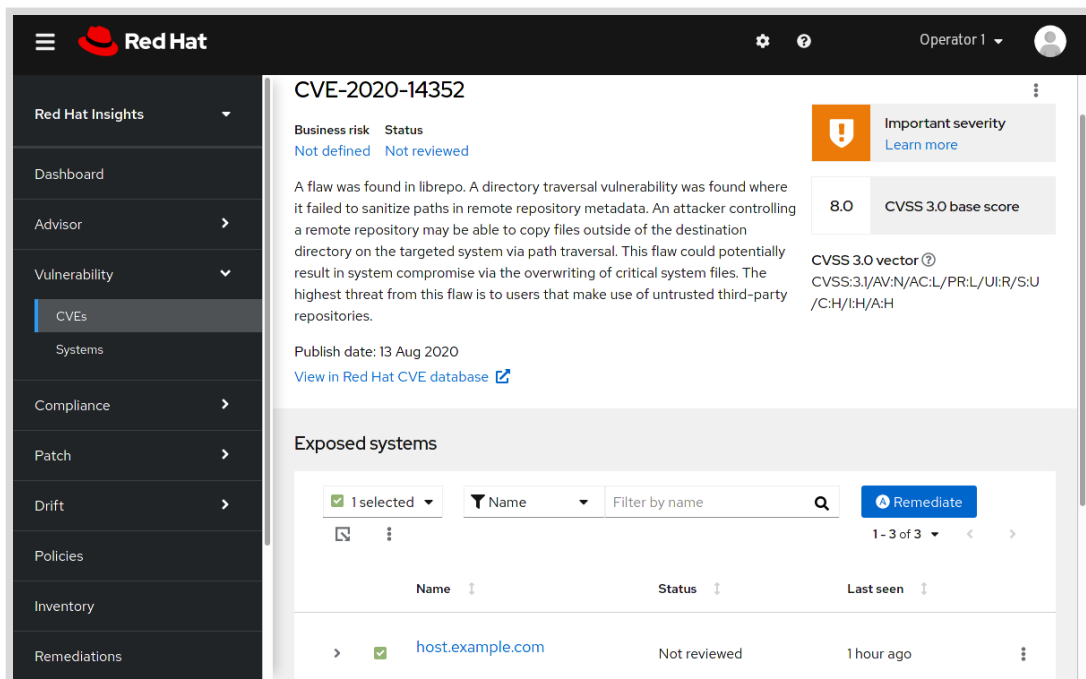


Figure 1.11: Details of a CVE

## Analyzing Compliance Using the Compliance Service

The Compliance service analyses your systems and reports their compliance level to an OpenSCAP policy. The OpenSCAP project implements tools to check the compliance of a system against a set of rules. Insights provides the rules to evaluate your systems against different policies, such as the *Payment Card Industry Data Security Standard (PCI DSS)*.

## Updating Packages Using the Patch Service

The Patch service lists the Red Hat Product Advisories applicable to your systems. It can also generate an Ansible Playbook that you can run to update the RPM packages associated with the applicable advisories. To access the list of advisories for a specific system, use the **Patch** → **Systems** menu. Click **Apply all applicable advisories** for a system to generate the Ansible Playbook.

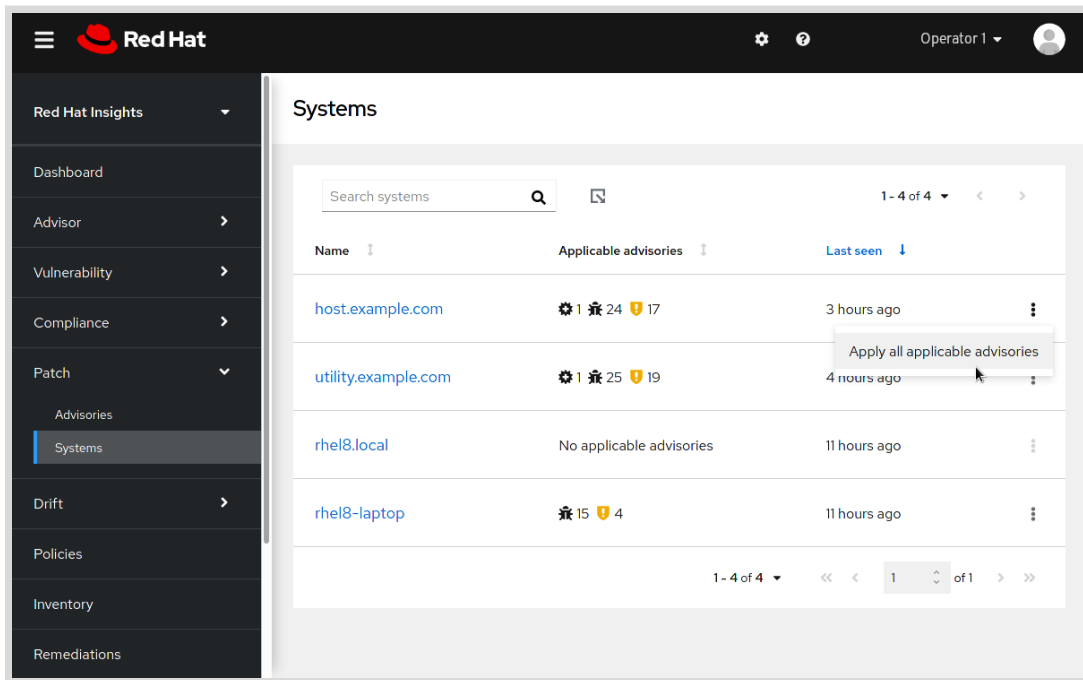


Figure 1.12: Patching a System

## Comparing Systems Using the Drift Service

Using the Drift service, you can compare systems, or a system history. This service can help you troubleshoot a system by comparing that system to a similar system, or to a previous system state. You access the service from the **Drift** → **Comparison** menu.

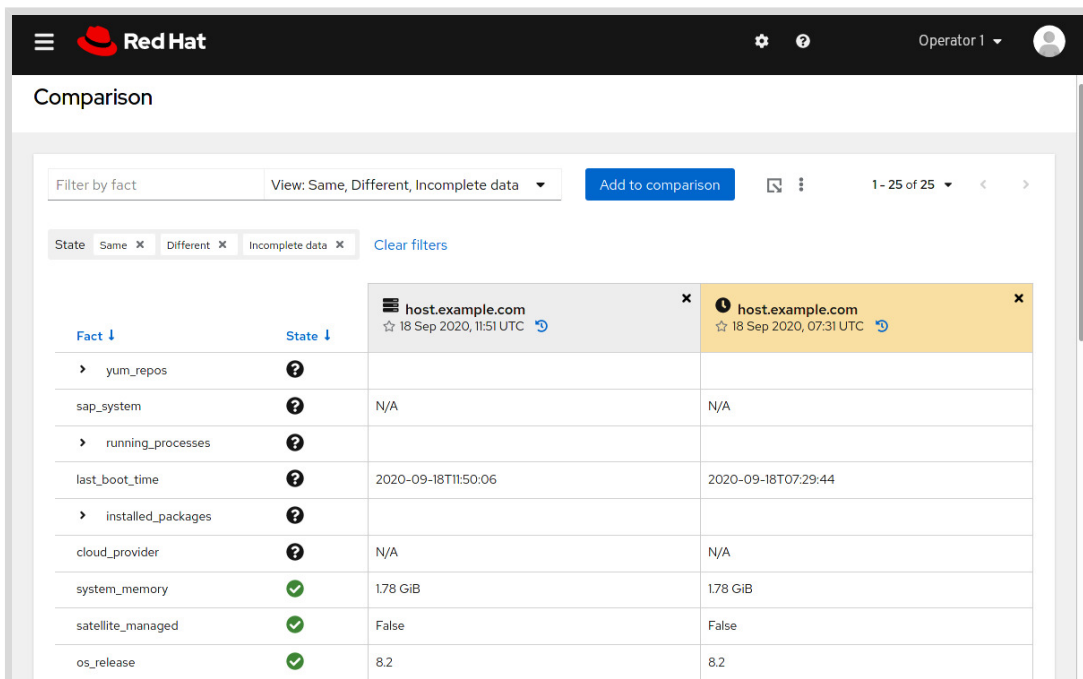


Figure 1.13: Comparing a System History

In the preceding screen capture, Insights compares the same system at two different times.

## Triggering Alerts Using the Policies Service

Using the Policies service, you create rules to monitor your systems and send alerts when a system does not comply with your rules. Insights evaluates the rules every time a system synchronizes its metadata. You access the service from the **Policies** menu.

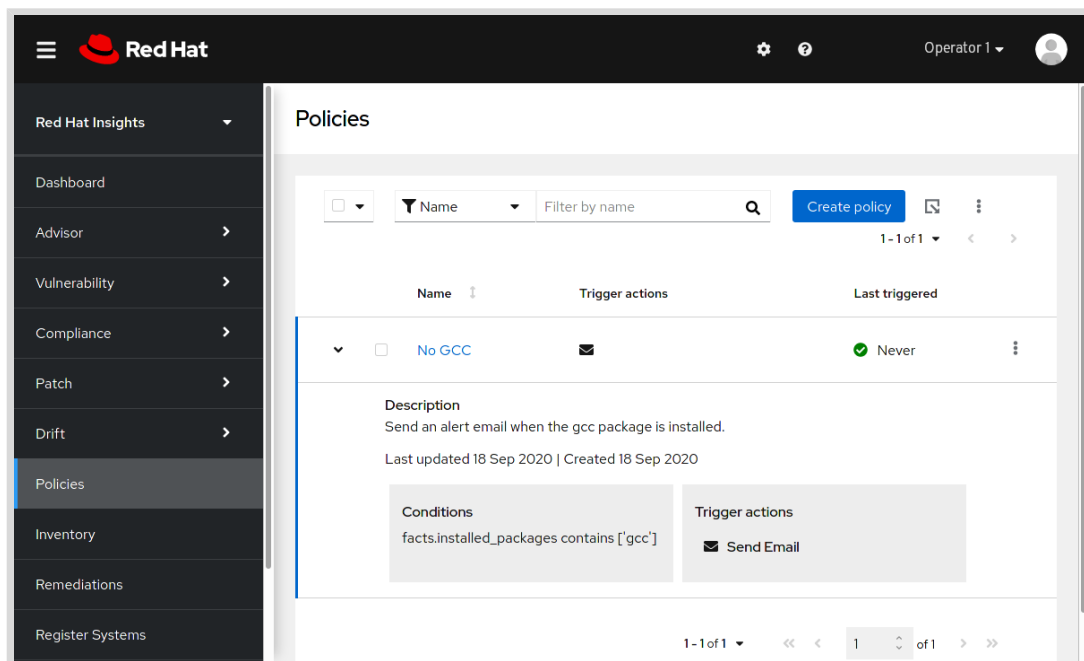


Figure 1.14: Details of a Custom Rule

## Accessing the Inventory and the Remediation Playbooks, and Monitoring Subscriptions

The **Inventory** page provides a list of the systems you have registered with Red Hat Insights. The **Last seen** column displays the time of the most recent metadata update for each system. By clicking a system name, you can review its details and directly access the Advisor, Vulnerability, Compliance, and Patch services for that system.

The **Remediations** page lists all the Ansible Playbooks that you created for remediation. You can download the playbooks from that page.

Using the **Subscription Watch** page, you can monitor your Red Hat subscription usage.



## References

**insights-client**(8) and **insights-client.conf**(5) man pages

For more information about Red Hat Insights, refer to the *Product Documentation for Red Hat Insights* at

[https://access.redhat.com/documentation/en-us/red\\_hat\\_insights](https://access.redhat.com/documentation/en-us/red_hat_insights)

For more information on excluding data collected by Insights, refer to the *Red Hat Insights client data obfuscation* and *Red Hat Insights client data redaction* chapters in the *Client Configuration Guide for Red Hat Insights* at

[https://access.redhat.com/documentation/en-us/red\\_hat\\_insights/2020-04/html-single/client\\_configuration\\_guide\\_for\\_red\\_hat\\_insights/index](https://access.redhat.com/documentation/en-us/red_hat_insights/2020-04/html-single/client_configuration_guide_for_red_hat_insights/index)

Information on the data collected by Red Hat Insights is available at

### **System Information Collected by Red Hat Insights**

<https://access.redhat.com/articles/1598863>

## ► Quiz

# Detecting and Resolving Issues with Red Hat Insights

Choose the correct answers to the following questions:

- 1. In what order do the following events occur when managing a Red Hat Enterprise Linux system using Red Hat Insights?
1. Red Hat Insights analyzes system metadata to determine which issues and recommendations apply.
  2. The Insights client uploads system metadata to the Red Hat Insights service.
  3. The administrator views the recommended actions in the Red Hat Insights customer portal.
  4. The Insights client collects system metadata on the Red Hat Enterprise Linux system.
- a. 1, 2, 3, 4  
b. 4, 2, 1, 3  
c. 4, 2, 3, 1  
d. 4, 1, 2, 3
- 2. Which command is used to register a client to Red Hat Insights?
- a. `insights-client --register`
  - b. `insights-client --no-upload`
  - c. `subscription-manager register`
  - d. `insights-client --unregister`
- 3. From which page in the Red Hat Insights console can you generate an Ansible Playbook to update the RPM packages on a system?
- a. **Advisor** → **Recommendations**
  - b. **Vulnerability** → **Systems**
  - c. **Patch** → **Systems**
  - d. **Remediations**

## ► Solution

# Detecting and Resolving Issues with Red Hat Insights

Choose the correct answers to the following questions:

- 1. In what order do the following events occur when managing a Red Hat Enterprise Linux system using Red Hat Insights?
1. Red Hat Insights analyzes system metadata to determine which issues and recommendations apply.
  2. The Insights client uploads system metadata to the Red Hat Insights service.
  3. The administrator views the recommended actions in the Red Hat Insights customer portal.
  4. The Insights client collects system metadata on the Red Hat Enterprise Linux system.
- a. 1, 2, 3, 4  
b. 4, 2, 1, 3  
c. 4, 2, 3, 1  
d. 4, 1, 2, 3
- 2. Which command is used to register a client to Red Hat Insights?
- a. `insights-client --register`
  - b. `insights-client --no-upload`
  - c. `subscription-manager register`
  - d. `insights-client --unregister`
- 3. From which page in the Red Hat Insights console can you generate an Ansible Playbook to update the RPM packages on a system?
- a. **Advisor** → **Recommendations**
  - b. **Vulnerability** → **Systems**
  - c. **Patch** → **Systems**
  - d. **Remediations**

## Summary

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In this chapter, you learned:

- The Bash shell is a command interpreter that prompts interactive users to specify Linux commands.
- Many commands have a **--help** option that displays a usage message or screen.
- SSH supports both password-based and key-based authentication.
- The **ssh-keygen** command generates an SSH key pair for authentication. The **ssh-copy-id** command exports the public key to remote systems.
- Red Hat Customer Portal provides you with access to documentation, downloads, optimization tools, support case management, and subscription and entitlement management for your Red Hat products.
- **redhat-support-tool** is a command-line tool to query Knowledgebase and work with support cases from the server's command line.
- Red Hat Insights is a SaaS-based predictive analytics tool to help you identify and remediate threats to your systems' security, performance, availability, and stability.